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To cite this article: Chian-Woei Shyu (2022) Energy poverty alleviation in Southeast Asian countries: policy implications for improving access to electricity, *Journal of Asian Public Policy*, 15:1, 97-121, DOI: [10.1080/17516234.2020.1770426](https://doi.org/10.1080/17516234.2020.1770426)

To link to this article: <https://doi.org/10.1080/17516234.2020.1770426>



Published online: 28 Jun 2020.



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Energy poverty alleviation in Southeast Asian countries: policy implications for improving access to electricity

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ABSTRACT

Over the past 20 years, the improvement of access to electricity in Southeast Asia has been substantial. The electrification rate in this region has improved from approximately 60% in 2000 to more than 70% in 2010 and more than 90% in 2018. However, there are still 45 million people without access to electricity living mainly in Cambodia and Myanmar where other poverty issues coexist. This study aims to review the progress of the improvement of access to electricity in this region and suggest policy implications for improving access to electricity to combat energy poverty.

ARTICLE HISTORY

Received 18 March 2020
Accepted 14 May 2020

KEYWORDS

Energy poverty; energy access; access to electricity; Southeast Asia

1. Introduction

Energy is essential for human life and development. Lack of modern energy services, particularly electricity, affects many aspects of human development and well-being at both the household level and the community level. The areas affected include household living quality, health and sanitation, children's education, economic development, women's empowerment, and social development. According to the energy data provided by the IEA (International Energy Agency) in 2018, 11% of the global population, representing approximately 840 million people, still lack access to modern electricity services (IEA, 2019a). The issue of energy poverty occurs mainly in developing countries. In 2018, the global electrification rate was 89%, whereas the electrification rate in developing countries was 86% (IEA, 2019a). Almost 97% of people who lack access to electricity live in developing countries. This shows the linkage between access to electricity and development level. The electrification rate is particularly low in the Least Developed Countries (LDCs). LDCs are defined by the United Nations (UN) as the low-income countries that are confronting severe structural impediments to sustainable development (United Nations, 2018). The electrification rate in LDCs in 2017 was only 51% (World Bank, 2020).

Besides the development level among countries, energy poverty in terms of access to electricity in rural areas also poses great challenges for developing countries. In 2018, the electrification rate in urban areas of the world and of developing countries reached 96% and 95%, respectively, whereas the electrification rate in rural areas of the world was 79% and that in developing countries was only 77% (IEA, 2019a).

The United Nations has increased its efforts to combat energy poverty. In 2015, the energy-related development issue was listed for the first time in the UN's development agenda, Sustainable Development Goals (SDG). SDG7, 'affordable and clean energy', aims to ensure access to affordable, reliable, sustainable, and modern energy services for all and considers energy to be central to nearly every major challenge and opportunity (United Nations, 2020). One of its three main targets, SDG7.1, is to ensure universal access to affordable, reliable, and modern energy services by 2030 (United Nations, 2020). The two indicators to measure the achievement of the SDG7.1 target are SDG7.1.1 'proportion of population with access to electricity' and SDG7.1.2 'proportion of population with primary reliance on clean fuels and technology' (United Nations, 2020). The proportion of the population with access to electricity listed as the first indicator of SDG7 shows the importance and priority requirement of access to electricity for human development.

In Asian developing countries, the electrification rate increased from 67% in 2000 to 79% in 2010 and to a peak of 94% in 2018 (IEA, 2019a). Compared to the electrification rate of 86% in developing countries worldwide, the electrification rate in Asian developing countries has been faster, particularly since 2010. The electrification rate in both rural and urban areas has also improved significantly. The electrification rates in urban and rural areas of Asian developing countries have reached 98% and 91%, respectively (IEA, 2019a). According to the available data by country in Asia from the IEA, the electrification rates remain lower than 80% in only four listed Asian developing countries: DPR Korea (27%), Myanmar (43%), Cambodia (72%), and Pakistan (77%) (IEA, 2019a). Two of these four countries are located in Southeast Asia. This study sets the scope of the analysis in Southeast Asia to draw more attention to and foster discussion on alleviation of energy poverty in terms of access to electricity in this region.

The objectives of this study are as follows: 1) review the concept of energy poverty, electricity access, living standards, and human development and the relationship among them; 2) review the energy profile in Southeast Asia and identify key features of energy patterns that are related to electricity access; 3) analyse the current status of access to electricity in Southeast Asia, focusing particularly on the lowest electrification-rate countries – Cambodia and Myanmar; 4) according to the results of the analysis, suggest policy implications for improving electricity access in this region to combat energy poverty and achieving SDG7, 'universal electricity access', by 2030.

The structure of this paper, besides introduction and conclusion section, is organized in five parts: 1) In section 'Methodologies', the author introduces which methods are applied in this study, how data are obtained, what kind of data are collected, how these data are analysed and discussed; 2) In section 'Energy poverty, electricity access, living standards and human development', the author reviews these concepts and discusses their relationship as well as their relevance; 3) In section 'Energy profile in Southeast Asia', the author first reviews this region's energy characteristics, energy resource endowments, energy structure, energy demand, energy consumption, electricity consumption and further elaborates some key features of the energy pattern related to electricity access; 4) In section 'Access to

electricity in Southeast Asia', the author presents the results of analysis in terms of access to electricity in this region as well as in comparison with the world, developing countries, developing countries of Asia with special focus on this region's lowest electrification-rate countries, Myanmar and Cambodia; 5) In section 'Policy implications for improving electricity access to combat energy poverty and achieve UN SDG7 universal electricity access by 2030', the author draws policy implications from the findings for improving access to electricity in this region to combat energy poverty.

2. Methodologies

The method applied in this study uses existing statistical data and literature. The energy data used for mapping the status of energy poverty in terms of access to electricity are from the IEA, UN, and World Bank databases, which provide more reliable, verified, and available data. The constraints of existing statistical data, instead of first-hand data, are the limitation of the form of data being used for analysis and the availability of other data and other forms of data. For example, some energy data for Lao People's Democratic Republic (Lao PDR) is from 2015, instead of 2018. The data do not provide the electrification rate of all developing countries in Asia. The validity and reliability of the data are unknown; however, IEA, UN, and the World Bank have been collecting and reporting energy data in terms of electrification rate for decades and are currently the most reliable and available open energy data providers.

In this study, due to the availability of data from the IEA database, the region of Southeast Asia refers exclusively to the following ten countries: Brunei, Cambodia, Indonesia, Lao PDR, Malaysia, Myanmar, Philippines, Singapore, Thailand, and Vietnam. East Timor is not included in this study. These ten countries are also the member states of the Association of Southeast Asian Nations (ASEAN).

The analysis of energy poverty in terms of access to electricity in Southeast Asia are based on the electrification rate and existing literature that focus on the discussions of access to electricity in Southeast Asian countries. The policy recommendations for improving access to electricity to combat energy poverty are drawn from and developed based on the analysis. The constraints of using second hand data and literature for the analysis in this study lie in the accuracy of the data and literature reflecting the reality. To complement the analysis of this study, the first-hand data collection such as interviews, observations, and field research are suggested to further verify the results of this study.

3. Energy poverty, electricity access, living standards and human development

3.1 Energy poverty and electricity access

Currently, there is no commonly agreed definition of the concept 'energy poverty'. There are different approaches and measurements to define and measure energy poverty (Khanna et al., 2019; Nussbaumer et al., 2012; Pachauri & Spreng, 2011). There exist

methodological challenges of different approaches and measurements when applying them to measure a country or a region's energy poverty (Herrero, 2017).

Energy poverty is defined by Foster et al. (2000) as the lack of availability of energy. Rao and Pachauri (2017) referred energy poverty to people who do not have access to energy. Modi et al. (2005, p. 9) defined energy poverty as 'the inability to cook with modern cooking fuels and the lack of a bare minimum of electric lighting to read, or for other household and productive activities after sunset'. Practical Action (2009, p. 1) defined energy poverty as 'the lack of adequate modern energy for the basic needs of cooking, warmth and lighting, and essential energy services for schools, health centers and income generation'. The UN secretary-general's Advisory Group on Energy and Climate Change (AGECC (The Secretary-General's Advisory Group on Energy and Climate Change), 2010, p. 8) defined energy poverty referring to the several billion people living in low-income countries 'who experience severe energy poverty in terms of inadequate and unreliable access to energy services and reliance on traditional biomass'. The above definitions of the concept 'energy poverty' refer to those living at the level of the poorest who lack access to electricity and clean fuel.

The definition of the concept 'energy poverty' also links to the concept of 'affordability' referring to household income. Boardman (1991) defined energy poverty to those households whose energy expenditure exceeds more than 10% of household income. Frankauser and Tepic (2007) and Bensch et al. (2010) defined energy poverty refers to those households who spend 10–30% of their household income on energy services. AGECC (2010, p. 13) suggested a reasonable cost of basic energy services should not be more than 10–20% of the household income. The above definitions of the concept 'energy poverty' refer to the affordability of energy services, assuming the households may already have access to electricity.

The concept 'energy poverty' discussed in this study refers to the concept of 'energy access' instead of only the concept of 'energy affordability', particularly to those people who do not have access to electricity or are not able to afford access to electricity. This study draw attention to the discussions on securing reliable, adequate and affordable access to electricity at household level to meet basic human needs.

3.2 Electricity access, living standards, and human development

Access to electricity links to living stands (or some alternative concepts, such as living quality, quality of life, well-being) and human development (UNDP (United Nations Development Programme), 2000; World Bank, 2001; Pachauri et al., 2004; Sagar, 2005; Bazilian et al., 2010; Barnes, 2011; Nussbaumer et al., 2012; Shyu, 2014; Rao & Pachauri, 2017; UNDP, 2019a; Sarkodie & Adams, 2020). At household level, having access to electricity improves living standards, e.g. increase access to news and information, improve learning conditions for children, increase entertainment, increase efficiency for domestic tasks, increase efficiency for income-generating tasks, increase personal security, increase family interaction, improve health and hygiene condition, save time for domestic tasks, save time for income-generating tasks, increase working hours for domestic tasks, increase working hours for income-generating tasks, increase flexibility of time for domestic tasks, increase flexibility of time for income generation tasks, etc. (Cook et al., 2005; ESMAP (Energy Sector Management Assistance Program), 2002; Riva et al., 2018;

United Nations, 2000; World Bank, 2000, 2008; Zhang et al., 2019). At community level, having access to electricity improves also household living standards, such as increase employment or job opportunities, increase small and micro rural business, improve health services, improve educational services, improve government services, etc. (Cook et al., 2005; Mapako & Mbewe, 2004; Riva et al., 2018; United Nations, 2000; World Bank, 2008; Zhang et al., 2019). The improvement of household living standards at household level and community level leads to improvement of living quality, quality of life, well-being, and ultimately human development. Access to electricity and the amount of electricity consumed by a household is hence a key contributor to human development (IEA, 2004) towards eradicating poverty (Rao & Pachauri, 2017).

4. Energy profile in Southeast Asia

Nearly one out of every ten people in the world live in this very densely populated region. Southeast Asia is a diverse and disparate region with huge differences in the scale of energy characteristics and energy resource endowments, energy structure, and patterns of energy demand and consumption, both among and within the countries.

4.1 Energy characteristics and energy resource endowments

Although the ten Southeast Asian countries are located in one geographical region, they have varied energy characteristics and energy resource endowments (see Table 1). Some countries have greater energy resource endowments to export, such as Brunei's export of oil and Liquefied Natural Gas (LNG), Malaysia's export of LNG (the world's third largest in 2018) and oil, and Indonesia's export of steam coal (the world's second largest in 2018) and LNG. Some countries have very limited energy resource endowment and rely heavily on energy import; these include Singapore (almost entirely dependent on energy imports), Thailand, and the Philippines (although the Philippines is the world's second-largest geothermal producer). Some countries, such as Myanmar and Cambodia, are featured for having a relatively lower electrification rate compared to other countries in this region.

4.2 Energy structure and energy demand

Southeast Asia's energy role is becoming increasingly vital in the overall world energy structure. This region accounts for nearly 5% of global energy demand, and this share has been increasing steadily in the past two decades (IEA, 2019b). Energy demand in Southeast Asia has increased by more than 80% since 2000, an average of 3.4% growth per year, far higher than the global annual average of 2.0% over the period (IEA, 2019b). With increasing population and rapid urbanization, the energy demand is expected to expand. Oil accounted for one third of primary energy demand in the energy structure of this region in 2018, followed by coal, gas, other renewables, traditional use of solid biomass, and hydro (IEA, 2019b). The energy structure of this region is fossil fuel-based, with more than 75% of the energy demand being met by oil, coal, and gas. According to IEA's forecasting, this trend will continue unchanged if no active energy policy interventions are implemented (IEA, 2019b). The share of oil in the energy structure is expected to decline, but the share of coal, gas, and renewable energy is expected to increase (IEA, 2019b).

Table 1. Key energy characteristics and energy resource endowments in Southeast Asia.

Country	Key energy characteristics and energy resource endowments	Sources
Brunei	Among the wealthiest countries in the world on a per-capita basis due to oil and LNG (Liquefied Natural Gas) exports. Significant producer and exporter of oil and gas. Gas is dominant fuel in electricity.	IEA (2013) IEA (2015)
Cambodia	Low level of electrification rate. Potential to develop oil and gas resources. Low levels of energy access and low per-capita energy demand. Potential for oil and gas production.	IEA (2013) IEA (2015)
Indonesia	Largest energy consumer in ASEAN (Association of Southeast Asian Nations), with massive scope for growth. It exports steam coal (the world's largest) and LNG, and is an increasing importer of oil. Largest energy consumer, rising domestic needs has led it to reorient some energy production to domestic market. World's largest coal exporter and major LNG exporter.	IEA (2013) IEA (2015)
Lao PDR	Aims to become the hydropower 'battery' of Asia. Electricity exports have been increasing sharply. Significant hydropower potential with aims to export to neighbouring countries.	IEA (2013) IEA (2015)
Malaysia	Third-largest energy consumer in ASEAN with relatively high per-capita consumption. It is an oil and LNG exporter. Third-largest energy consumer. World's second-largest LNG exporter in 2014. Oil exporter, but not a net producer.	IEA (2013) IEA (2015)
Myanmar	Abundant hydropower and natural gas resources. The development of hydropower and natural gas resources is vital to reduce poverty and support economic growth. Low levels of energy access and high reliance on biomass and hydropower. Increasing domestic gas production.	IEA (2013) IEA (2015)
Philippines	Fast rising electricity demand requires expanded supplies. Strongly reliant on energy imports. It is the world's second-largest geothermal producer. Heavily dependent on imports. World's second-largest geothermal power producer.	IEA (2013) IEA (2015)
Singapore	It has become Asia's key oil trading and refining hub (the third largest in the world) and could become a major gas hub. Key global refining and petrochemical hub and developing gas-trading hub. Almost entirely dependent on energy imports.	IEA (2013) IEA (2015)
Thailand	Second-largest energy consumer in ASEAN and heavily dependent on energy imports due to limited energy resources. It aims to diversify electricity generation. Second-largest energy consumer. Oil and gas producer, but increasingly dependent on imports. Gas is primary fuel in electricity.	IEA (2013) IEA (2015)
Vietnam	Significant renewable and fossil fuel resources, but rapidly growing energy demand underlies a shift towards imports. It is developing a nuclear power programme. Growing energy demand has led to increasing imports despite domestic production of fossil fuels. Making progress towards introduction of nuclear power.	IEA (2013) IEA (2015)

Source: Compiled by author based on IEA (2013, p. 2015).

As for energy demand per capita, the ten countries in this region vary significantly and show very great gaps. According to the data of IEA (2019b) energy demand per capita in Brunei, Singapore, Malaysia, and Thailand in 2017 is above world average. However, the energy demand per capita in the remaining six countries is relatively low by comparison with international data. The energy demand per capita in Brunei and Singapore even exceeds that of OECD (Organization for Economic Co-operation and Development) countries and is more than six times than that in Indonesia and the Philippines, more than 10 times than that in Lao PDR and Vietnam, and more than 12 times than that in Myanmar and Cambodia. This also shows that the scope for further demand growth in this region is expected to be enormous.

4.3 Energy consumption and electricity consumption

Since 2000, oil for transport and electricity for building have together accounted for almost 50% of the increase in final energy consumption in Southeast Asia (IEA, 2019b). The increase of oil and electricity consumption is related closely to the broader trend of rapid urbanization and growth in middle-class households, which results in increased demand for mobility and the use of electrical appliances (IEA, 2019b).

As for electricity consumption, the region's average annual growth is among the world's fastest, at 6% per year since 2000, only behind China (10%) and India (7%) (IEA, 2019b). According to IEA's estimation, there is still large scope for use of electricity to grow in overall energy consumption. As for electrical appliances, the electricity demand for residential space cooling has been particularly strong due to ownership of air conditioners almost tripling, resulting in the associated electricity consumption quadrupling (IEA, 2019b). The need for space cooling is expected to be continually high as only 15% of households in this region have air conditioners compared with almost 80% in Singapore and Malaysia (IEA, 2019b).

The expansion in electricity demand has been largely met by a rise in coal-fired power generation, followed by gas and hydropower (IEA, 2019b). Among different renewable energy sources, hydropower has dominated the expansion of renewable energy-based power, particularly in Vietnam, Malaysia, Indonesia, Lao PDR, Myanmar, and Cambodia (see Figure 1). Renewable energy is expected to play an increasingly vital role in electricity generation; however, without stronger policy support and efforts, renewable energy can meet only approximately one third of the increased power generation needs (IEA, 2019b).

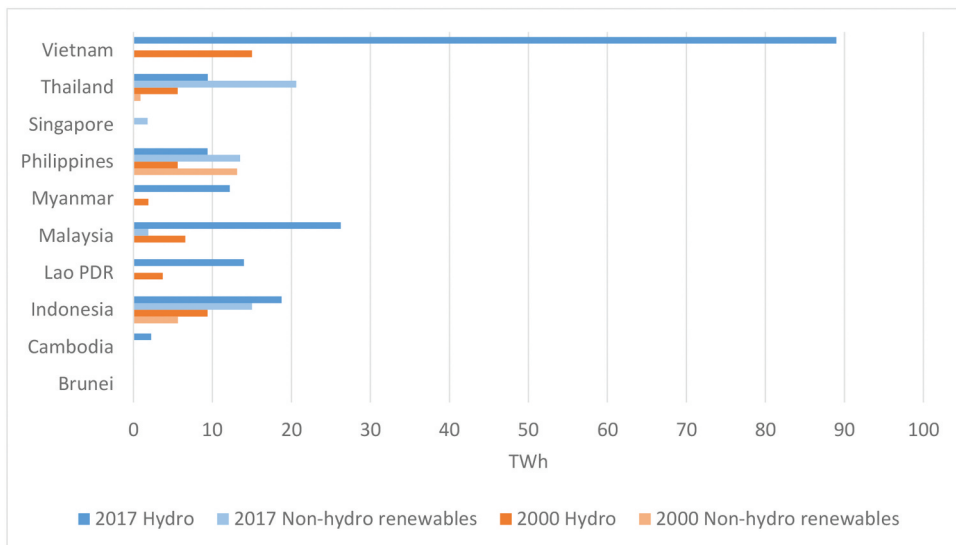


Figure 1. Electricity generation from renewable energy sources in Southeast Asia. *Source:* Compiled by author based on IEA (2019b, p. 33). *Note:* 1. TWh refer to terawatt-hours; 2. Data for Lao PDR is 2015.

4.4 Key features of the energy pattern related to electricity access

Based on the above data and analysis, this study summarizes some key features of the energy pattern related to the focus on ‘access to electricity’.

4.4.1 Increasing electricity consumption and demand

This region is expected to have a significant increase in electricity consumption and demand due to economic and demographic factors. The annual economic growth in this region has averaged approximately 5–6% over the past two decades (IEA, 2019b). This has resulted in rising incomes, an emerging new consumer class, and accelerating urbanization. In 2018, nearly 50% of Southeast Asia’s population lived in urban areas, whereas less than 40% of Southeast Asia’s population lived in urban areas in 2000 (IEA, 2019b). Each year, the region’s urban population increases by approximately seven million people (IEA, 2019b). The increasing electricity consumption and demand will further enlarge the gap of development and poverty between members of the population both with and without access to modern electricity services.

4.4.2 Increasing air pollution, public health risks, climate risks, and environmental degradation

The increasing electricity consumption and demand has brought significant costs in terms of air pollution, public health risks, climate risks, and environmental degradation. This regions’ response to the rising electricity consumption and demand has focused mainly on increasing the supply of fossil fuel, particularly coal, which is less sensitive to price fluctuations and does not add burdens for consumers in terms of affordability (IEA, 2019b). The use of coal for power generation, the use of oil for transportation, and the use of biomass for the residential sector are the three main energy-related sources of pollutant emissions that cause poor air quality that threatens human health (IEA, 2019b). Approximately 80% of the population in this region is exposed to different levels of fine particulate matter that exceed the minimum air quality standard defined by the World Health Organization (IEA, 2019b). The use of biomass for cooking and heating in the residential sector not only threatens human health due to indoor pollution but also causes environmental degradation. The increasing fossil fuel consumption has led to a doubling of regional energy-related CO² emissions in the past two decades and further increased global climate risks (IEA, 2019b). The current population without access to electricity is particularly more vulnerable to these risks.

4.4.3 ASEAN plays a vital role in improving access to electricity

ASEAN Secretariat plays a vital role in terms of institutional body in energy policy and providing intra-regional co-operation platform in this region. Several long-standing and intra-regional cooperation energy projects, such as the ASEAN Power Grid and the Trans-ASEAN Gas Pipeline, have been implemented collectively with the aim of establishing interconnected grids for electricity and natural gas among member states in this region (IEA, 2013). The 4th ASEAN Plan of Action on Energy Cooperation (APAEC), covering the period 2016–2020, is expected to focus on promoting increased interconnections between national power grids to enable cross-border power trade, developing and deploying renewable energy sources, improving energy efficiency, and co-operating on

politically contested ‘clean coal technologies’, e.g. CO² capture and storage (CCS), reflecting on the increasing dependence on coal and the efforts to reduce pollutants and CO² emissions in this region (IEA, 2015). Apart from that, a pilot project for the electricity trade, which involves a transit from Lao PDR through Thailand and Malaysia to Singapore is currently being implemented (IEA, 2015). The interconnection of electricity grids across borders and the development of electricity trade mechanisms under the ASEAN framework play an important role in improving access to electricity for the remaining population without access to electricity.

5. Access to electricity in Southeast Asia

5.1 Southeast Asia in comparison with the world and developing countries

In 2018, the world electrification rate reached 89% (see Table 2). The proportion of the world’s population without access to electricity has fallen sharply, from 51% in 1970 to 41% in 1990, 27% in 2000, 20% in 2010, and 11% in 2018 (IEA, 2002, 2004, 2019a). Approximately 96.6% of the population without access to electricity lives in developing countries, and 79% lives in rural areas. The electrification rates in Central and South America, developing Asia, and Middle Eastern areas have all exceeded 93%. The remaining population without access to electricity is located mainly in Africa and rural areas. Compared with the electrification rate in the world (89%) and in developing countries (86%), the electrification rate in Southeast Asia (90%) is slightly higher.

The most severe situation is in Africa, where the electrification rate is 54%, and particularly in Sub-Saharan Africa, where the electrification rate is only 45%. The electrification rate in rural areas of Sub-Saharan Africa is lowest, at 26%. Compared to other regions, the improvement of access to electrification in rural areas of this region remains very slow.

5.2 Southeast Asia in comparison with developing countries of Asia

The electrification rate in most parts of Asia has improved dramatically in the past two decades and has reached more than 90% (see Table 3). However, some countries have electrification rates lower than 90%, including Bangladesh (85%), Pakistan (77%), Cambodia (72%), other Asia (63%), Myanmar (42%), and DPR Korea (27%). Compared with the electrification rate in other Asian developing countries (94%), whether in urban areas (98%) or rural areas (91%), the electrification rate in the region of Southeast Asia (90%) is slightly lower overall and also in urban (91%) and rural areas (83%). In addition, the gap in the electrification rate between urban and rural areas of this region is wider than those in Asian developing countries. The electrification rate in urban areas of this region is almost 97%, whereas the electrification rate in rural areas remains at approximately 83%.

5.3 Electrification rate in Southeast Asian countries

Millions of people in the region have gained access to electricity since 2000. The electrification rate has improved from approximately 60% in 2000 to more than 70% in 2010, and rapidly to more than 90% in 2018 (see Table 4). Some 45 million people in the region are still without access to electricity (IEA, 2019b). Access to electricity varies widely across the region in terms

Table 4. Electrification rate in Southeast Asia.

Area/Country	National					Urban	Rural
	2000	2005	2010	2015	2018	2018	2018
Indonesia	53%	56%	67%	88%	98%	>99%	96%
Other Southeast Asia	65%	76%	79%	85%	90%	97%	83%
Brunei	>95%	>95%	>95%	>95%	>95%	>95%	>95%
Cambodia	<5%	12%	23%	49%	72%	>95%	64%
Lao PDR	43%	57%	75%	91%	95%	>95%	93%
Malaysia	>95%	>95%	>95%	>95%	>95%	>95%	>95%
Myanmar	5%	13%	24%	30%	43%	82%	26%
Philippines	74%	77%	80%	90%	>95%	>95%	95%
Singapore	>95%	>95%	>95%	>95%	>95%	>95%	>95%
Thailand	82%	>95%	>95%	>95%	>95%	>95%	>95%
Vietnam	76%	95%	>95%	>95%	>95%	>95%	>95%

Source: Compiled by author based on IEA data from the IEA (2019a). *Access to electricity database*. Retrieved 5 February 2020 from <https://www.iea.org/reports/sdg7-data-and-projections>.

of countries and urban/rural areas. There also wide disparities in income and development levels, leaving areas where progress in access to electrification has lagged behind.

In 2018, the electrification rate in eight of the ten Southeast Asian countries has exceeded 95%. Some countries, including Brunei, Malaysia, Singapore, Thailand, and Vietnam, have had a have high electrification rate above 95% since 2005. The remaining six countries have seen a very significant improvement in their electrification rates since 2000; i.e., Cambodia (67% increase from less than 5% to 72%), Lao PDR (52% increase from 43% to 95%), Indonesia (45% increase from 53% to 98%), Myanmar (38% increase from 5% to 43%), and the Philippines (20% increase from 75% to 95%).

The electrification rate is lower than 95% in just two countries: Myanmar (43%) and Cambodia (72%). Although the electrification rate in urban areas of Cambodia reaches 95%, the electrification rate in rural areas remains at 64%. The electrification rate in urban areas of Myanmar reaches 82%, which is lower than that in the other urban areas in this region, at above 95%. The lowest electrification rate (26%) in this region is found in rural areas of Myanmar. Overall, the population without access to electricity is distributed in parts of the urban areas of Myanmar, rural areas of Cambodia and Myanmar, and other remote rural areas of this region.

5.4 Policies and targets for improving access to electricity in Southeast Asia

Aside from Brunei, Malaysia, Singapore, and Thailand, the other six countries in this region have energy polices and targets aimed at improving access to electricity (see Table 5).

Vietnam, Indonesia, the Philippines, and Lao PDR plan to achieve universal access to electricity by 2025. Vietnam aims to ensure 100% electrification of rural households by 2020. Besides the major electricity source, hydropower, the country plans to increase the share of non-hydro renewables-based generation capacity to 12.5% by 2025 and 21% by 2030, which will contribute to the improvement of access to electricity in remote rural areas where no hydropower is available. The Philippines aims to achieve 100% electrification by 2022. The country plans to expand the grid to interconnect all major islands by increasing the electricity generation capacity from 16 GW in 2011 to 29 GW in 2030 and triple the installed capacity of renewables-based power generation from the 2010 level to 15 GW by 2030, with most of the growth coming from geothermal and hydropower. Indonesia aims to achieve

Table 5. Targets and policies related to the targets for improving access to electricity in Southeast Asia.

Country	Targets	Policies related to the targets	Sources
Brunei	N/A	N/A	IEA (2019b, 2017a) IEA (2013)
Cambodia	Electrification for all villages by 2020 70% electrification for households by 2030 Increase grid-quality electricity access to 70% of households by 2030 100% electricity access in any form for villages by 2020	Increase hydropower capacity to 2241 megawatts by 2020 Develop hydropower to lower cost of power domestically	IEA (2019b, 2017a) IEA (2013)
Indonesia	Achieve electrification ratio of 99.7% by 2025 Boost electricity access to 99% of households by 2020	N/A N/A	IEA (2019b, 2017a) IEA (2013)
Lao PDR	Achieve electrification rate of 98% by 2025 Increase household electrification to 80% by 2015 and 90% by 2020	N/A Build 5 GW of new hydropower capacity and 1.9 GW of coal-fired capacity by 2015 Improve transmission lines in the northern, central and southern areas and links with Thailand and Vietnam	IEA (2019b, 2017a) IEA (2013)
Malaysia	N/A	N/A	IEA (2019b, 2017a) IEA (2013)
Myanmar	Achieve electrification rate of 80% by 2030 Achieve total electrification rate of 100% by 2030	N/A N/A	IEA (2019b) IEA (2017a)
Philippines	N/A Achieve 100% electrification by 2022 Increase household electrification rate from 70% to 90% by 2017 and 100% sitio ('small township') electrification by 2015	N/A Triple the installed capacity of renewables-based power generation from 2010 level to 15 GW by 2030 Increase generation capacity from 16 GW in 2011 to 29 GW in 2030 and expand grid to interconnect all major islands Triple installed renewable capacity to 15 GW in 2030, with most of the growth from geothermal and hydropower	IEA (2013) IEA (2019b, 2017a) IEA (2013)
Singapore	N/A	N/A	IEA (2019b, 2017a) IEA (2013)
Thailand	N/A	N/A	IEA (2019b, 2017a) IEA (2013)
Vietnam	Ensure most rural households have access to electricity by 2020 Reach 100% electrification of rural households by 2020	Increase the share of non-hydro renewables-based generation capacity to 12.5% by 2025 and 21% by 2030 N/A	IEA (2019b, 2017a) IEA (2013)

Source: Compiled by author based on IEA (2013, 2017a, 2019b).

electrification rates of 99% by 2020 and 99.7% by 2025. Lao PDR aims to achieve an electrification rate of 98% by 2025.

Compared to the above four countries, Cambodia and Myanmar have less ambitious targets for improving the electrification rate. Cambodia aims to achieve 100% electrification for villages by 2020 but 70% electrification for households by 2030. The main energy source for improving access to electricity is grid-based electricity powered by hydro. Myanmar aims to achieve an electrification rate of 80% by 2030.

Although national policies contribute to the improvement of electrification rates in this region, the rapid urbanization may be the actual factor driving the increase in the electrification rate. Each year, approximately seven million people move from rural areas to urban areas in the region (IEA, 2019b). This migration contributes to reducing the number of people without access to electricity. Unless there are more national policies aimed at improving access to electricity in remote rural areas, achieving the UN SDG7, 'universal electricity access', by 2030 will remain unlikely, particularly in Myanmar and Cambodia.

5.5 Focus on Southeast Asia's lowest electrification-rate countries: Myanmar and Cambodia

Access to electricity links economic development and human development (IEA, 2010, 2017b). Currently, Myanmar and Cambodia have the lowest electrification rates in Southeast Asia. The targets for improving the electrification rate by 2030 are not set to achieve the UN SDG7, 'universal electricity access'. In 2018, these two countries, together with Lao PDR, were listed among the 47 countries in the UN's Least Developed Country' category (see Table 6).

Table 6. LDC status in 2018.

LDC ^a info	Cambodia	Lao PDR	Myanmar	LDCs	Developing countries	Inclusion ^b	Graduation ^c
Inclusion year	1991	1971	1987	N/A	N/A	N/A	N/A
Gross national income (GNI) per capita	\$1,075	\$1,996	\$1255	\$1,229	\$7,064	\$1,025 or below	\$1,230 or above
Human assets index (HAI) ^d	68.9	72.8	68.5	53.1	76.4	62.0 or below	66.0 or above
Economic vulnerability index (EVI) ^e	34.8	33.7	31.7	41.3	34.7	36.0 or above	32.0 or below
Graduation eligibility	No	1 st time in 2018	1 st time in 2018	N/A	N/A	N/A	N/A

Source: Compiled by author based on UN (2018).

^aThree criteria are used for identifying LDCs: gross national income (GNI) per capita; human assets index (HAI) and economic vulnerability index (EVI).

^bTwo criteria for inclusion eligibility: a) inclusion threshold must be met for all three criteria (per capita GNI, HAI, EVI) at one triennial review; b) population must be less than 75 million.

^cTwo criteria for graduation eligibility: a) graduation threshold must be for two of the three criteria (per capita GNI, HAI, EVI) at two consecutive triennial reviews; b) alternatively, per capita GNI is at least twice the graduation threshold (income-only criterion).

^dComposition of HAI: a) health index (1/2): percentage of population undernourished (1/6), under-five mortality rate (1/6), maternal mortality rate (1/6); b) education index (1/2): gross secondary school enrolment ratio (1/4), adult literacy rate (1/4).

^eComposition of EVI: a) exposure index (1/2): population (1/8), remoteness (1/8), merchandise export concentration (1/16), share of agriculture, hunting, forest and fishing (1/16), share of population in low-elevated coastal zones (1/8); b) shock index: instability of exports of goods and services (1/4), victims of natural disasters (1/8), instability of agriculture production (1/8).

In terms of human development, the Human Development Index (HDI) of both Myanmar and Cambodia is the lowest among the ten countries (see Table 7). In terms of poverty, the Multidimensional Poverty Index (MPI) of both Myanmar and Cambodia is the highest among the ten countries (see Table 7). Energy-related indicators, electricity, and cooking fuel are two of the 10 indicators that make up MPI. More than 20% of the population in both countries is vulnerable to multidimensional poverty (see Table 7).

Based on IEA (2019b) historical data of the average annual growth in per-capita electricity consumption in this region from 2000 to 2015, this growth is of positive relevance to the average annual growth in per-capita GDP, particularly in Myanmar, Cambodia, and Vietnam. The data show the economic growth in terms of GDP linked to the growth of electricity consumption. This implies that the population gaining access to electricity during this period has increased their consumption dramatically and further leads to the improvement of their economic development and presumably also their human development.

The above data show the connection between access to electricity, economic development, and human development. Whether improved access to electricity is either the cause or a consequence of the improvement of economic development and human development, improved access to electricity doubtless helps to improve economic development and human development. Hence, improving access to electricity is one of the

Table 7. Human Development Index (HDI) in 2018 and Multidimensional Poverty Index (MPI) status.

HDI ^a info	Category	HDI	HDI rank	MPI ^b	Population vulnerable to multidimensional poverty
Brunei	Very high human development	0.845	43	N/A	N/A
Cambodia	Medium human development	0.581	146	0.170 (2014)	21.1%
Indonesia	High human development	0.707	111	0.028 (2017)	9.1%
Lao PDR	Medium human development	0.604	140	0.108 (2017)	21.2%
Malaysia	Very high human development	0.804	61	N/A	N/A
Myanmar	Medium human development	0.584	145	0.176 (2016)	21.9%
Philippines	High human development	0.712	106	0.024 (2017)	7.3%
Singapore	Very high human development	0.935	9	N/A	N/A
Thailand	High human development	0.765	77	0.003 (2016)	7.2%
Vietnam	Medium human development	0.693	118	0.019 (2014)	5.6%
Developing countries	N/A	0.686	N/A	0.114	15.3%
Least developed countries	N/A	0.528	N/A	N/A	N/A
World	N/A	0.731	N/A	N/A	N/A

Source: Compiled by author based on UNDP (2019a, 2019b, 2019c).

^aComposition of HDI: Life expectancy index, education index, GNI index.

^bComposition of MPI: a) health: Nutrition (1/6), children mortality (1/6); b) education: years of schooling (1/6), school attendance (1/6); c) Standard of living: electricity (1/18), sanitation (1/18), drinking water (1/18), housing (1/18), cooking fuel (1/18), assets (1/18).

crucial factors in alleviating poverty. To cope with the issue of access to electricity, the governments of Myanmar and Cambodia have taken different policy actions.

5.5.1 Policy actions to improve access to electricity in Myanmar

In Myanmar, the government has initiated the National Electrification Plan and set targets to improve access to electricity. The improvement of access to electricity relies mainly on grid-based electricity provided by the government with a centralized approach. However, a large number of people live in rural areas that are relatively far from the grid. In remote and rural areas, the government have initiated different PPP (Public Private Partnership) programs to improve access to electricity. In cooperation with the international aid organization, the World Bank, the government provided 400 USD million financial support for rural electrification programmes to provide off-grid and mini-grid systems to reach 500,000 households by 2021 to improve the application of a renewable energy-based off-grid electricity supply in rural areas (World Bank, 2015). To improve the affordability of electricity use for poor households, the government initiated a 'pay-as-you-go' financing scheme using a PPP approach to introduce private investment from private sector for the poorest to pay for a very small amount of electricity, e.g. four dollars a month worth, to power two fluorescent lamps (IEA, 2015). A Singapore-based pay-as-you-go provider SolarHome estimated that they have provided more than 40,000 homes access to electricity in rural areas of Myanmar in 2018 (Renewable Energy World, 2018). Through scratch cards or mobile payments, SolarHome provided off-grid households with a solar lighting system with an initial 10 USD down payment, followed by daily, weekly, or monthly repayments, and plans to extend its services to systems with bundled television sets (Renewable Energy World, 2018).

5.5.2 Policy actions to improve access to electricity in Cambodia

In 2004, the Cambodia Ministry of Mines and Energy established a Rural Electrification Fund (REF), financed by the World Bank and the Asian Development Bank (ADB), to provide interest free loans or grants for private electricity suppliers to improve access to electricity and poor households to improve the affordability of electricity (Electricity Authority of Cambodia, 2016; IEA, 2013). The improvement of access to electricity relies mainly on grid-based electricity from hydropower and coal (IEA, 2017a). Electricity suppliers have to obtain licences from the Cambodia Electricity Authority to supply electricity; however, no provinces are covered fully by the licenced electricity suppliers (IEA, 2017a). The government has increased licenced electricity suppliers in the past decade. In 2016, the REF contributed to extending the transmission lines for private electricity suppliers, with more than 2,800 km in rural areas (IEA, 2017a). The development of solar photovoltaic (PV) small grid systems in rural areas of Cambodia is at a very early stage. The first large-scale solar PV installation, with 10 MW capacity, was introduced in 2017 with the cooperation with the ADB (Asian Development Bank), 2017).

Another key issue to improve access to electricity in rural areas of Cambodia is the affordability of electricity use. The electricity tariffs in 2015 in Cambodia were higher than were the average residential electricity tariffs in Indonesia, Malaysia, Thailand, and Vietnam (Electricity Authority of Cambodia, 2016). In addition, electricity tariffs in the provinces with lower electrification rates are higher than are those in the provinces with higher electrification rates (IEA, 2017a). For poor households, the REF initiated a 'Power to

the Poor' programme to provide consumers in rural areas with interest-free loans for in-house wiring and equipment and connection fees and deposits required by private electricity suppliers (IEA, 2017a). The REF has also facilitated the development of solar home systems and micro- and mini-hydropower systems in rural areas (IEA, 2013). In rural areas of Cambodia, small diesel electricity generators are dominant, but the price is costly compared to grid-based electricity tariffs (Electricity Authority of Cambodia, 2016). Since 2008, the German development bank KfW has financed the 'Power to the Poor' programme to provide solar home systems for rural households (IEA, 2017a). In 2016, more than 10,000 solar home systems were installed under the aegis of this international cooperation (IEA, 2017a).

6. Policy implications for improving electricity access to combat energy poverty and achieve UN SDG7 'universal electricity access' by 2030

Based on the analysis in 'Access to electricity in Southeast Asia', the current energy poverty in terms of access to electricity in Southeast Asia is to be found in parts of urban areas of Myanmar, rural areas of Cambodia and Myanmar, and some remote rural areas of this region. There is also great potential and possibility for this region to achieve the UN SDG7 'universal electricity access' by 2030 if more policy interventions and financial resources are put in place. To further improve access to electricity in this region, this study suggests the following policy implications.

6.1 Grid-based electricity using hydropower: a more stable, effective, economical, and sustainable way to improve access to electricity in this region

Grid-based electricity provides a more stable, reliable, and effective electricity supply and shall be considered as the first priority in improving access to electricity in this region, particularly in urban areas in Myanmar and less remote rural areas of this region. According to IEA's estimation, access to electricity in the coming decade in this region will mainly be from grid-based electricity, but this will account only for approximately 40% by 2030 (IEA, 2019b). This study suggests that the share of grid-based electricity should be increased.

Based on the analysis in section 'Energy consumption and electricity consumption', 'Increasing electricity consumption and demand', and 'Increasing air pollution, public health risks, climate risks, and environmental degradation', the expansions in electricity demand are largely met by coal power plants and gas power plants, which will increase air pollution, damage public health, and exacerbate climate risks in this region. Besides coal and gas, generation of electricity from renewables has been led mainly by hydropower in the Greater Mekong Sub-region. The development and transboundary management of hydropower in the Greater Mekong Sub-region (GMS) has historically involved political, geopolitical, economic, and social, and environmental factors in this region among five ASEAN members (including Cambodia, Lao PDR, Myanmar, Thailand, and Vietnam) plus China (Sneddon & Fox, 2006; Thiesmeyer, 2012; Intralawana et al., 2018). Since 1992, the ADB has brought these six countries together to launch GMS development program for political, geopolitical, economic cooperation, particularly the development of hydropower, electricity trade, and a sub-regional market with interconnections (IEA, 2013, 2015). There are considerable untapped potential hydropower and expansion of hydropower infrastructures although increasing

environmental and social challenges have made the development difficult (Thiesmeyer, 2012; IEA, 2013, 2015, 2017a; Intralawana et al., 2018). China has been also actively invested in the development of hydropower projects in this Sub-region, particularly in Cambodia, Lao PDR, and Myanmar (IEA, 2017a) which have the lowest electrification rate in this region. The development of hydropower in this Sub-region provides great potential through regional cooperation to meet the electricity demand of the remaining population without access to electricity in this region.

This study suggests that hydropower will be a more sustainable energy source, with less air pollution, fewer public health risks, and fewer climate risks, in meeting the electricity demand of the remaining population without access to electricity in this region. Compared with coal and gas, hydropower generated locally is also a more economical option in terms of energy price and less transmission loss to generate electricity. With more cross-border electricity interconnections from countries in this region, such as Lao PDR and Vietnam, with abundant hydropower to countries with less hydropower, in addition to increased trade of hydropower-based electricity between these countries, hydropower can play a vital role in the future improvement of access to electricity in this region.

6.2 Stand-alone mini-grids and stand-alone small home systems using renewable energy: a more feasible way to improve access to electricity in 'remote' rural areas of this region

In remote rural areas of this region, or on some islands in the Philippines and Indonesia where grid-based electricity is either difficult or uneconomical to supply, stand-alone mini-grids and stand-alone small home systems based on renewable energy, such as solar, hydro, and wind can improve access to electricity, particularly in Myanmar, where many remote rural areas are not connected to the grid. These two types of infrastructure involve different types of potential suppliers.

Given limited government funding, PPP project financing and governance are suggested as an adequate model to enhance efficiency, fairness, legitimacy and accountability of development projects in the context of developing countries (Oinarov et al., 2017). Stand-alone mini-grids are usually financed by the government or PPP involving international aid organizations (e.g. the World Bank, Asian Development Bank, or other national development bank such as German development bank KfW) and/or private suppliers due to its larger scale and investment. Stand-alone mini-grids are usually run by the communities or/and private suppliers. Stand-alone small home systems however are much smaller scale, usually to power two or three lamps and other low Watt electric appliances, such as TV, radio, and mobile phone. Stand-alone small home systems are easy to operate by a household. Stand-alone small home systems are usually financed by the government or PPP involving international aid organizations (e.g. the World Bank, Asian Development Bank, or other national development bank such as German development bank KfW) and/or partially/completely individual household.

This study suggests that grid-based electricity will have to be the first priority in increasing the electricity supply and distribution, whether the electricity is generated either from national grids or stand-alone mini-grids using indigenous renewable energy or from stand-alone hybrid mini-grids using renewable energy and diesel. People tend to abandon stand-alone small home energy systems when they are connected to grid. To

avoid wasted investment and ensure a long-term electricity supply, grid-based electricity will have to be the main electricity supply in remote rural areas.

The improvement of access to electricity with stand-alone small home systems using renewable energy will have to be restricted to 'very remote' rural areas where neither stand-alone mini-grids based on indigenous renewable energy nor hybrid mini-grids using renewable energy and diesel can supply electricity. Unlike hydropower, the development of solar PV and wind power in this region is just beginning (IEA, 2019b). The levelised cost of electricity in Southeast Asia has declined by 65% for solar PV over the past five years, which has stimulated deployment in some countries of this region (IEA, 2019a). Together with small hydropower systems outside the Mekong River Basin and wind energy, renewable energy offers great potential for improvement of access to electricity in remote rural areas of this region.

6.3 National government's policy commitments and interventions: key to improving access to electricity in this region

Practical Action (2016) in its *Poor People's Energy Outlook 2016* criticized one of the inadequacies of energy access plan today is not keeping the needs of those without access to electricity at the centre of the policy and planning agenda. The historical data of energy characteristics and energy resource endowments in section 'Energy structure and energy demand', policies and targets for improving access to electricity in section 'Policies and targets for improving access to electricity in Southeast Asia', and the development level in section 'Focus on Southeast Asia's lowest electrification-rate countries: Myanmar and Cambodia' show considerable differences in political direction, economic development, and energy resource endowments among the ten countries in this region. These data reveal a significant relationship between the ambitiousness of national governments' policy targets for improving access to electricity and the actual improvement of the electrification rate. Some countries have adopted or announced more ambitious policy targets and measures to improve access to electricity, such as Indonesia, Philippines, Vietnam and Lao PDR. And the improvements of electrification rates in these four countries are significant. Cambodia and Myanmar, which have the lowest electrification rates, have less ambitious policy targets and measures to improve access to electricity. Lao PDR, which has a similar development level to that of Cambodia and Myanmar in terms of LDC status, HDI, and MPI, has improved its electrification rate to above 95% in both urban and rural areas. Some successful electrification programs, such as China's Township Electrification Program, Indonesia's natural gas-based electrification plans, indicate strong determination and favourable policy to mobilize national resources and push forward electrification programs at the local level is the key to improve the rate of electrification to the level of developed countries (ESMAP, 1996; IEA, 2002, 2017b; Yang, 2003; Shyu, 2012). More policy commitments and interventions will have to be in place in both countries to alleviate energy poverty in terms of access to electricity. Energy sector reforms may affect the progress of energy access programs (Practical Action, 2018). Hence, energy sector reforms are also suggested to provide a better foundation to facilitate the process of the improvement of access to electricity.

6.4 Policy interventions focusing on financing mechanism: key to improving affordability for the rural and poor households in this region

Policy interventions should particularly address the financing mechanism to support the affordability of electricity use for the rest of the population that is not yet connected to electricity. Based on the analysis of LDC status, HDI, and MPI in Cambodia and Myanmar, the population living in remote or rural areas without access to electricity very likely also suffers poverty issues and requires great extent of financial support from the government to be able to afford electricity use. With the constraints of limited government financial resources in countries like LDC, PPP involving international aid organizations and private sectors providing a 'stable' funding for the remote and rural poor plays a key role to ultimately improve access to electricity in remote and rural areas. Risks may occur for the remote and rural poor to be disconnected to electricity due to unstable funding or short-term funding.

Different financing mechanisms and instruments are suggested by the Word Bank (2013), IRENA (2016) and Practical Action (2017) for expanding energy access: Grants (capital grants, gap funding, convertible subsidies, research & development grants), debt finance (subordinate debt, convertible loan, concessional loan, senior debt, micro-financing, on-lending), equity finance (long term equity, venture capital), guarantees & insurance (debt guarantees, risk insurance, foreign exchange), asset-backed securities (bonds, bundling), and results-based financing (output based disbursements, carbon financing, contingent development grants). Ortiz et al. (2007) suggested different financial instruments for off-grid electrification, including Rural electrification assistance (subsidy, donations, and discounts), funds (national or international funds, grants, partnerships), micro-financing (credit, leasing, fee for service), fiscal instruments (tax exemption or reduction, bulk tendering), and other mechanisms (quotas, feed-in tariffs, green certificates). Bhattacharyya (2013) differentiated project-level financing (including equity or debt financing by government, asset-based lending, non-recourse financing, supplier credits) and end-use level financing approaches (small-scale lending, micro credit, leasing arrangements, revolving funds). This study suggests that the focus of financing should be end-user oriented, particularly those without access to electricity. For grid based or mini-grid based projects, the systems need to be funded by the government whether the funding are from the government, PPP with international organization bilaterally or multilaterally, or PPP with private sectors. The electricity price from grid or mini-grid should be subsidized by government along with other financial mechanism such as pay as you go, small-scale lending, micro credit, leasing arrangements, revolving funds either from private sectors or government. For small home electricity systems, such as stand-alone solar home systems, government needs to provide subsidies for households along with some financial support from small-scale lending, micro credit, leasing arrangements, revolving funds either from private sectors or government.

6.5 Electricity interconnection, integration, and trade across borders: enhance and speed up access to grid-based electricity in this region

Based on the analysis in section 'ASEAN plays a vital role in improving access to electricity', the current electricity interconnection, integration, and trade in this region comes mainly under the framework of ASEAN. The ASEAN Power Grid plan initiated in 1997 plays several roles in the region's energy security, including expanding access to electricity, and has made

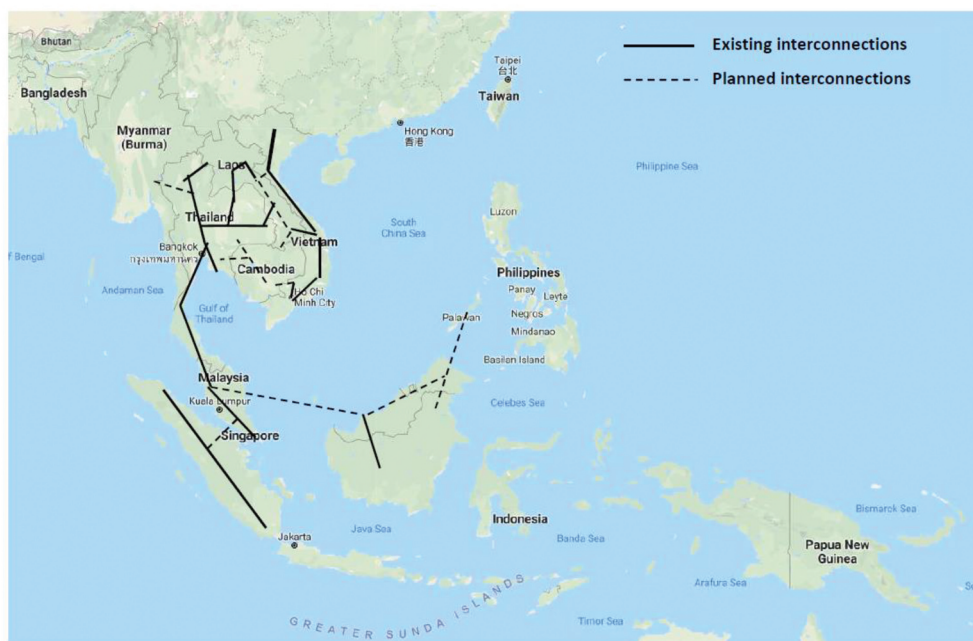


Figure 2. Southeast Asia existing and planned power grid interconnections. Source: Compiled by author based on IEA (2019b, p. 138); Google (n.d.).

significant progress in developing the ASEAN Power Grid (see Figure 2) (IEA, 2019b). But the electricity interconnections in this region have not been extended widely in Myanmar which is located on the far northwest edge of the electricity interconnection network. The developments of electricity interconnection, integration, and trade, particularly in hydropower, have been primarily bilateral (IEA, 2019b). More cooperation needs to be established at multilateral level, for example, hydropower development in Lao PDR was supported by increased export trading with the neighbouring countries, Thailand, Cambodia, Malaysia, and Vietnam (IEA, 2019b).

For geographical reasons, Cambodia has electricity interconnections with and imports electricity from Lao PDR and Vietnam, which have abundant hydropower (see Figure 2 and Table 8). The geographical advantages make it easier to extend the electricity grid to urban and rural areas in Cambodia than in Myanmar. Currently, Lao PDR and Thailand have electricity interconnections with rural areas of Myanmar that facilitate the improvement of access to electricity. However, the country has great potential for the development of electricity interconnections, integration, and trade with the other ASEAN member states to improve its electrification rate.

To achieve UN SDG7 'universal electricity access' by 2030 in this region, the ASEAN member states need to further address the issue of access to electricity to alleviate energy poverty and improve the quality of life of the population in rural areas of this region through ASEAN's policy framework. More work is necessary, particularly the multilateral cooperation between Myanmar government and other ASEAN member states, to establish multilateral trading, broader electricity system integration, and greater institutional development to further improve access to electricity. As broader geopolitical perspective, China's geopolitical

Table 8. Electricity interconnections and trading in Southeast Asia.

Country	Interconnections with which country	Export to which country	Import from which country	Bidirectional, non-financial exchange with which country
Brunei	None	None	None	None
Cambodia	Lao PDR Thailand Vietnam	Thailand	Lao PDR Thailand Vietnam	None
Indonesia	Malaysia ^a	Malaysia	None	None
Lao PDR	Cambodia Myanmar Thailand Vietnam	Cambodia Malaysia Thailand Vietnam	None	None
Malaysia	Indonesia* Singapore*	None	Lao PDR Thailand	Singapore
Myanmar	Thailand Lao PDR	None	None	None
Philippines	None	None	None	None
Singapore	Malaysia	None	None	Malaysia
Thailand	Cambodia Lao PDR Malaysia Myanmar	None	Cambodia Lao PDR	None
Vietnam	Cambodia Lao PDR	Cambodia	Lao PDR	None

Source: Compiled by author based on IEA (2019b).

^aElectricity interconnections are used primarily for security and back up.

interests in investment in the development of hydropower projects in the Greater Mekong Sub-region, particularly in the lowest electrification-rate countries like Cambodia, Lao PDR, and Myanmar, have effects on the improvement of electricity access. The ambition to integrate China's transportation infrastructure between Yunnan Province, Guangxi Zhuang Autonomous Region (GZAR) of the mainland and this region under Belt and Road Initiative (BRI) poses potential spillover effects to facilitate Myanmar's connectivity in terms of energy. However, the political and economic influences of China in this region will also bring other political and economic unexpected effects and risks as well.

Besides the framework of ASEAN, international cooperation with other countries, international organizations, and international development financial institutions also provides immense opportunities. The UN SDG7 targets 7.A and 7.B are (1) 'by 2030, enhance international cooperation to facilitate access to clean energy research and technology, including renewable energy, energy efficiency and advanced and cleaner fossil-fuel technology, and promote investment in energy infrastructure and clean energy technology' (Target 7.A) (United Nations, 2020) and (2) 'by 2030, expand infrastructure and upgrade technology for supplying modern and sustainable energy services for all in developing countries, in particular least developed countries, small island developing States, and land-locked developing countries, in accordance with their respective programmes of support' (Target 7.B) (United Nations, 2020). These two targets both offer and facilitate potential international financial resources to attract investment in energy structure and expand energy services to improve access to electricity in rural areas of this region, particularly for Myanmar and Cambodia.

7. Conclusions

Compared with other regions, the improvement of access to electricity in Southeast Asia over the past 20 years has been substantial and provides electricity for millions of people in this region. The alleviation of energy poverty in terms of access to electricity has been significant. Although rapid urbanization contributes to the improvement of the electrification rate in this region, the efforts made by the governments also facilitate the process. As of 2020, the remaining population that still suffers energy poverty in terms of access to electricity in this region are found in parts of urban areas in Myanmar, rural areas in Cambodia and Myanmar, and other remote rural areas of this region, and constitutes 45 million people. These areas also have other poverty issues. With the improvement of access to electricity and increasing electricity consumption in this region, the gaps in development and quality of life between areas with and without access to electricity further increase and worsen. To alleviate energy poverty in terms of access to electricity, national governments play a crucial role in triggering domestic policy interventions and international cooperation. Different financial mechanisms to support the poor households are particularly essential to further improve the electrification rate for the remaining population. This study suggests that grid-based electricity using renewable energies, particular hydropower, should be the first priority to resolve access to electricity in this region. For the very remote areas that are either impossible or uneconomical to connect to the grid, stand-alone small home systems based on indigenous renewable energy should supplement the needs of the remaining un-electrified areas in this region. For future studies, the research design using field research, household interview, and big data (if available) to investigate or monitor the progress on the improvement of access to electricity on the ground will provide valuable data for scientific community in understanding and advancing knowledge of energy poverty as well as contribute to government policy interventions, designs, and actions to combat energy poverty.

Acknowledgments

The author acknowledges the valuable comments and contributions given by the reviewers to improve the quality of this article. The author also acknowledges the support from editors, Prof. Qin Gao (Columbia University) and Prof. Alex Jingwei He (The Education University of Hong Kong), to select this paper to be part of this special issue 'Poverty alleviation in Asia: A comparative perspective' at *Journal of Asian Public Policy*.

Disclosure statement

No potential conflict of interest was reported by the author.

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